The Double Case - Sine Rule

CASE:

Triangle ABC is such that

AB = 9 cm

BC = 6 cm

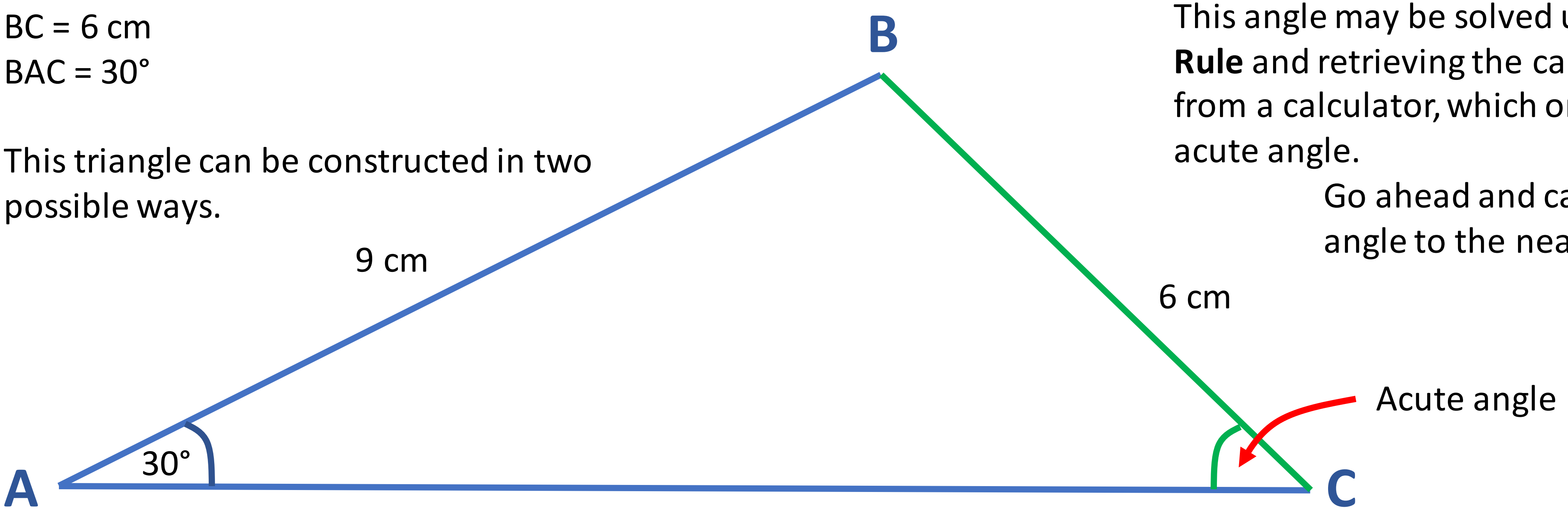
BAC = 30°

This triangle can be constructed in two possible ways.

The first way is with ACB having an **acute angle** (less than 90°).

This angle may be solved using the **Sine Rule** and retrieving the calculated value from a calculator, which only provides the acute angle.

Go ahead and calculate the acute angle to the nearest degree!



Required knowledge

Sine rule: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ **OR** $\frac{a}{\sin A} = \frac{b}{\sin B} =$

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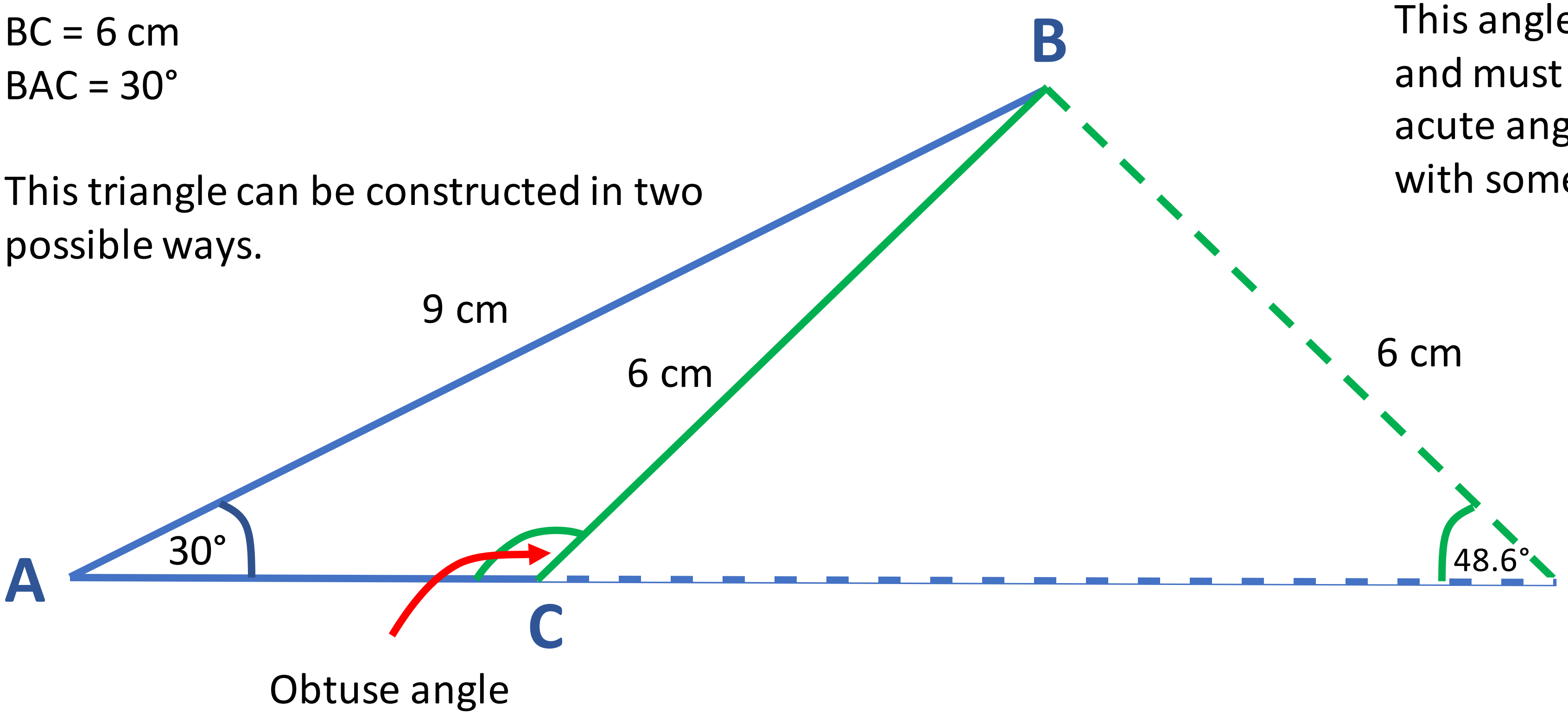
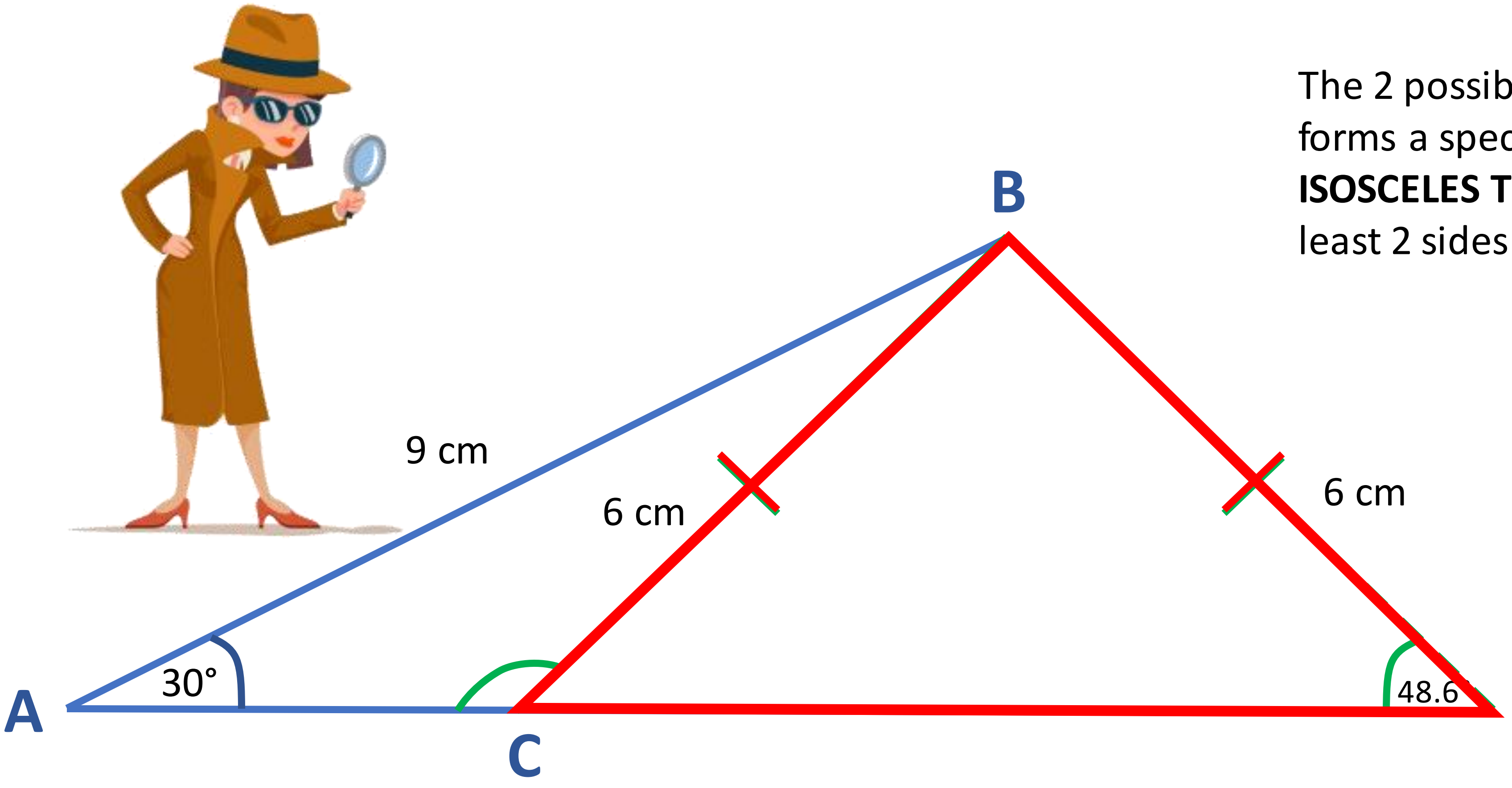


The second way is with ACB having an **obtuse angle** (more than 90°).

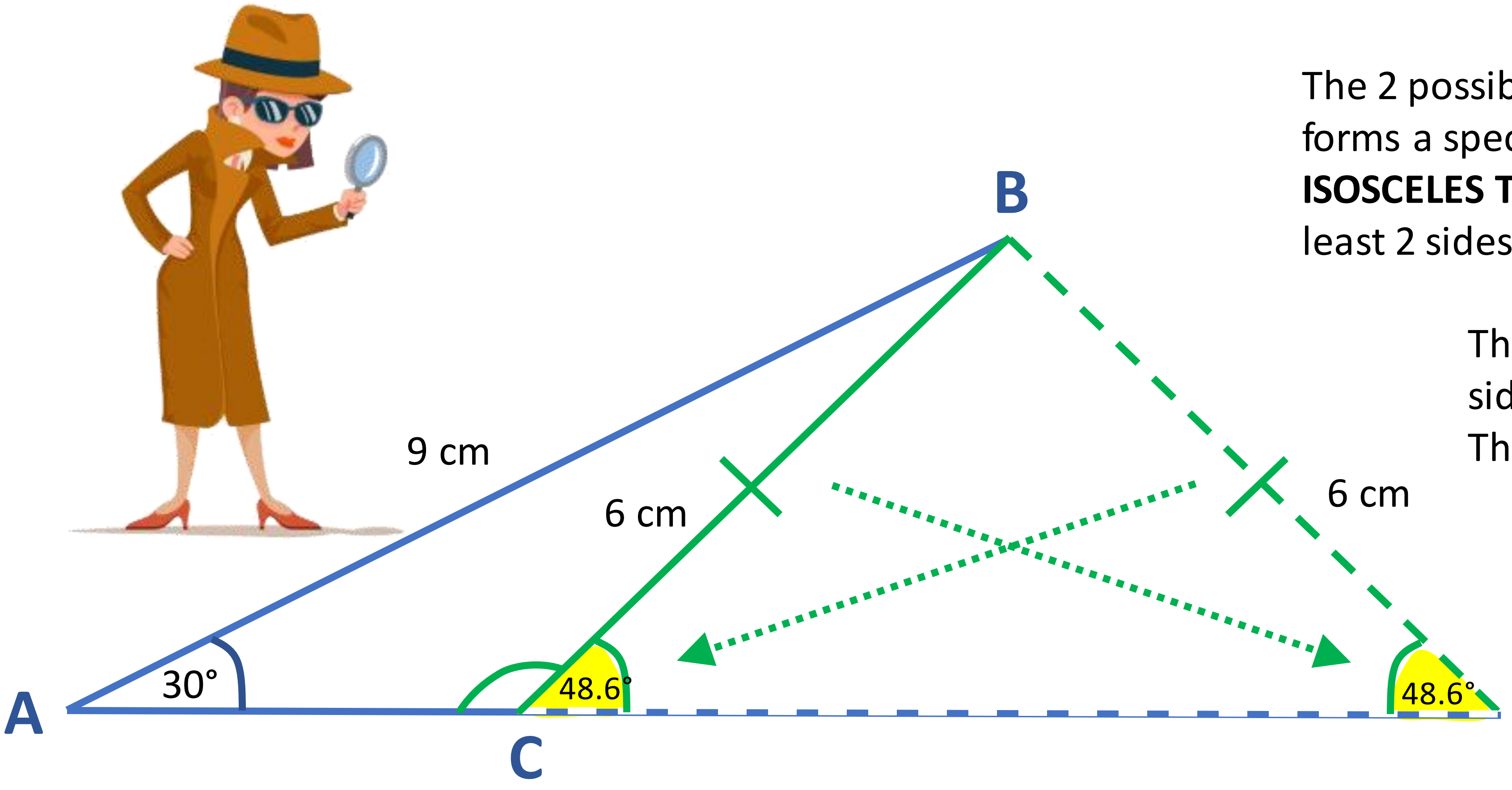
This angle is **NOT** given by the calculator, and must be determined by using the acute angle previously calculated along with some mathematical detective work.



The 2 possible orientations of line BC forms a special type of triangle called an **ISOSCELES TRIANGLE** because it has at least 2 sides of equal length.

Required knowledge

Isosceles triangle: A triangle with two equal sides. The angles the equal sides are also equal.



The 2 possible orientations of line BC forms a special type of triangle called an **ISOSCELES TRIANGLE** because it has at least 2 sides of equal length.

The **angles opposite** to the equal sides **are also equal**.
These are called **base angles**.

Therefore value of the acute angle (previously calculated) is next to the obtuse angle on a straight line.

Required knowledge

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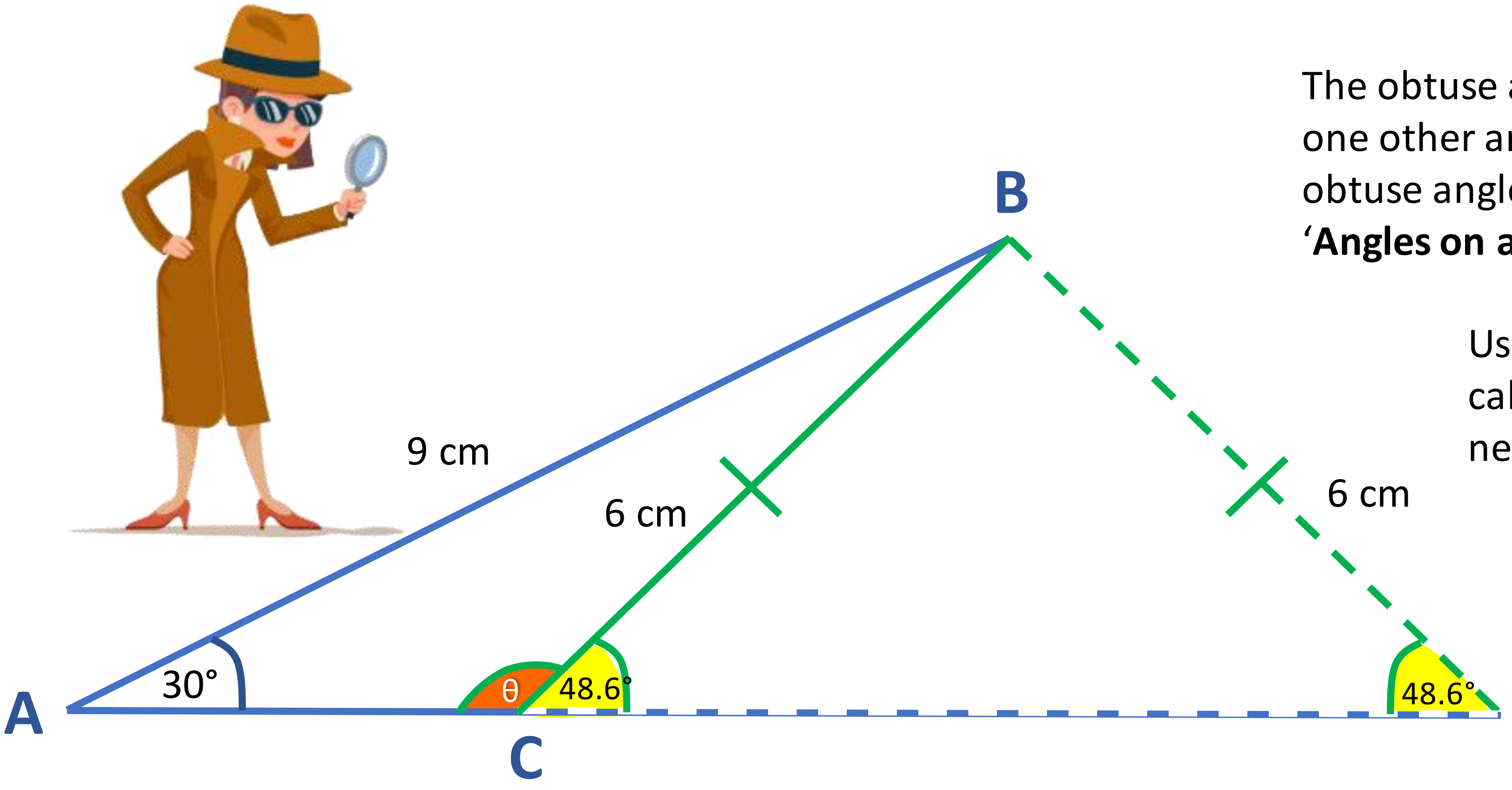


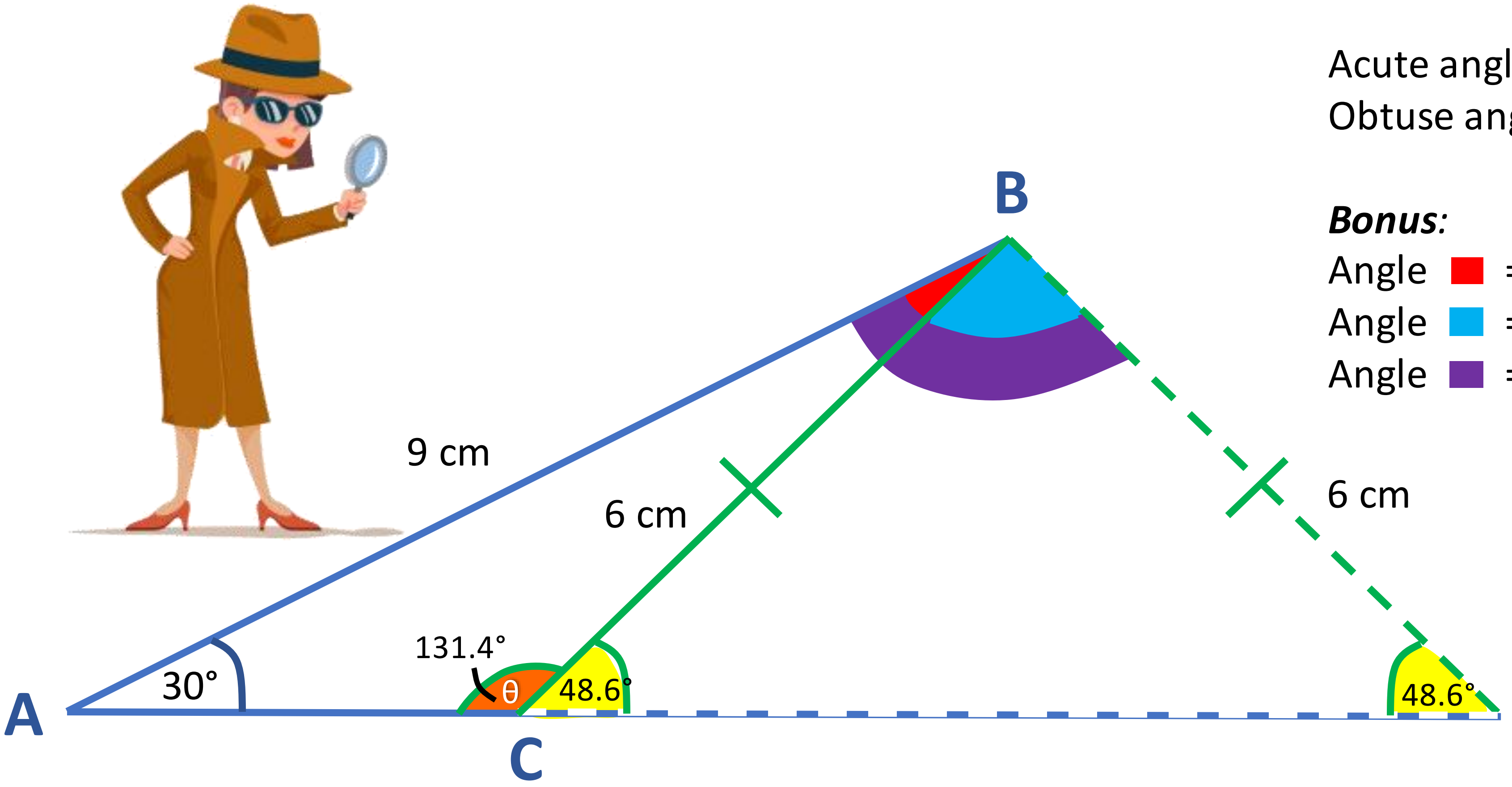
The obtuse angle lies on a straight line with one other angle of 48.6°, therefore the obtuse angle can be determined using the ‘**Angles on a Straight Line**’ rule.

Using this rule, Go ahead and calculate the obtuse angle to the nearest degree!

Bonus:
Do you know another rule that can help you determine the 3rd angle in each triangle we drew today?

Required knowledge

Angles on a Straight Line rule: Angles on a straight line add up



Calculations

Acute angle = $\sin^{-1}$ (sin(30) *9/6) = 48.6°

Obtuse angle = 180° – 48.6° = 131.4°

Bonus:

Angle (red) = 180° – (30° + 131.4°) = 18.6°

Angle (blue) = 180° – (48.6° + 48.6°) = 82.8°

Angle (purple) = 180° – (30° + 48.6°) = 101.4°

Required knowledge

Angles in a Triangle: Angles in a triangle add up to 180°.